

Seat No.: _____

No. of Printed Pages: 02

[68]

SARDAR PATEL UNIVERSITY

BCA (Semester 3) Examination October-2024

US03AEBCA06 (Discrete Mathematics)



Date: 21/10/2024 (Monday)

Time: 02:00 PM to 03:30 PM

Total Marks: 25

(Q 1) Multiple Choice Questions.

[05]

- 1 Let $u = (2024, 0, 1)$ and $v = (1, 0, -2024)$ then $u \cdot v =$ _____
A) 2024 B) 0 C) 1 D) -2024
- 2 If matrix $M = \begin{pmatrix} 0 & x \\ -2024 & 0 \end{pmatrix}$ is skew symmetric then $x =$ _____
A) 0 B) -2024 C) 2024 D) 1
- 3 The norm of $u = (-3, 4, -5, 5, -5)$ is,
A) 10 B) -5 C) 4 D) 75
- 4 The diameter of complete graph K_{2024} is,
A) 2024 B) 1012 C) 0 D) 1
- 5 In a connected map with $V = 20, E = 28$ then $R =$ _____
A) 50 B) 8 C) 46 D) 10

(Q 2) Let $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 0 & -2 \\ -1 & -2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 5 & -1 \\ 5 & 3 & 0 \\ -1 & 0 & 4 \end{pmatrix}$ then find

[10]

(i) $A^T + B^T$

(ii) $\det(A + B)$

(iii) $(A - B)^T$

(iv) Is it true that $(A - B)^T = A^T + B^T$? Justify.

OR

(1)

C.P. T. O.)

(Q 2) (i) Define: Orthogonal matrix

[10]

(ii) Show that the matrix $A = \frac{1}{3} \begin{pmatrix} 2 & 4 & 1 \\ 0 & 1 & 4 \\ 2 & 7 & -2 \end{pmatrix}$ is orthogonal.

(Q 3) (i) Define: Complete graph

[10]

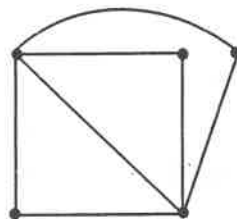
(ii) Draw a diagram of complete graph K_5 and K_6 .

(iii) Find the number of edges in complete graph K_{11} and K_{200} .

OR

(Q 3) Find incidence matrix and adjacency matrix for the given graph.

[10]



— X —